

Relations and Functions

- **Definition 1**

Let X be a set. A **relation** $\sim$ on X is a rule that determines for every $x, y \in X$ whether the statement

$$x \sim y$$

is true or false. If it is true, we say that x is related to y . If it is false, we write $x \not\sim y$.

- **Example 1.1**

Let $X = \{1, 2, 3, 4\}$, the following are examples of relations on X .

1. The relation $\sim$ given by $1 \sim 2, 2 \sim 3, 3 \sim 1, 4 \sim 4$.
2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 2 \sim 4$.
3. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.

- **Definition 2**

Let X be a set and $\sim$ a relation on X .

1. The relation is **reflexive** if $x \sim x$ for every $x \in X$.
2. The relation is **symmetric** if for every $x, y \in X$

$$x \sim y \implies y \sim x.$$

3. The relation is **antisymmetric** if for every $x, y \in X$

$$x \sim y \wedge y \sim x \implies x = y.$$

4. The relation is **transitive** if for every $x, y, z \in X$

$$x \sim y \wedge y \sim z \implies x \sim z.$$

- **Example 2.1**

On the set $\mathbb{Z}$ consider the relation

$$x \sim y \iff x \leq y.$$

Note that

1. $\sim$ is reflexive since $x \leq x$ for every $x \in \mathbb{Z}$.
2. $\sim$ is not symmetric since $1 \leq 2$ but $2 \not\leq 1$.
3. $\sim$ is antisymmetric because if $x \leq y$ and $y \leq x$ then $x = y$.
4. $\sim$ is transitive since $x \leq y$ and $y \leq z$ implies that $x \leq z$.

- **Definition 3**

Let X be a set. An **equivalence relation** on X is a relation $\sim$ that satisfies the following properties.

1. $\sim$ is reflexive.
2. $\sim$ is symmetric.
3. $\sim$ is transitive.

- **Example 3.1**

Let $X = \{1, 2, 3, 4\}$. The following are examples of equivalence relations on X .

1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 2 \sim 3, 3 \sim 2, 1 \sim 4, 4 \sim 1$.

- **Definition 4**

Let X be a set. An **order relation** on X is a relation $\sim$ that satisfies the following properties.

1. $\sim$ is reflexive.
2. $\sim$ is antisymmetric.
3. $\sim$ is transitive.

◦ **Example 4.1**

Let $X = \{1, 2, 3, 4\}$. The following are examples of order relations on X .

1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 1 \sim 2, 1 \sim 3, 1 \sim 4, 2 \sim 3, 2 \sim 4, 3 \sim 4$.

• **Definition 5**

A **function** $f: X \rightarrow Y$ is a correspondence of elements of X with elements of Y such that for all $x \in X$ there exists a unique $y \in Y$ such that $y = f(x)$.

$$\begin{aligned} f: X &\rightarrow Y \\ x &\mapsto y = f(x) \end{aligned}$$

The element $y = f(x)$ is called the **image** of x under f and the element x is called the **preimage** of y under f .

• **Definition 6**

If $f: X \rightarrow Y$ is a function, the **domain** of f denote $\text{Dom}(f)$ is X and the **codomain** of $\text{CoDom}(f)$ is Y . The **range** of f , denoted $\text{Ran}(f)$, is the set

$$\text{Ran}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$$

Note that $\text{Ran}(f) \subseteq Y$.

◦ **Example 6.1**

1. The function $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$.
2. The function $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sin(x)$.
3. The function $h: [0, \infty) \rightarrow \mathbb{R}, h(x) = \sqrt{x}$.

◦ **Example 6.2**

Let $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$ the following examples of functions from X to Y .

1. The function $f: X \rightarrow Y$ defined by

$$f(1) = a \quad f(2) = b \quad f(3) = c.$$

2. The function $g: X \rightarrow Y$ defined by

$$g(1) = a \quad g(2) = a \quad g(3) = a.$$

3. The function $h: X \rightarrow Y$ defined by

$$h(1) = a \quad h(2) = b \quad h(3) = a.$$

• **Definition 7**

Two functions $f: X \rightarrow Y$ and $g: Z \rightarrow W$ are **equal** if $X = Z, Y = W$ and $f(x) = g(x)$ for all $x \in X$.

◦ **Example 7.1**

1. Let X be a set. The **identity function** on X is the function $\text{id}_x: X \rightarrow X$ given by $\text{id}_x(x) = x$.
2. Let X and Y be sets and $c \in Y$. The **constant function** with value c is the function $K_c: X \rightarrow Y$ given by $K_c(x) = c$.
3. Let X be a set and $A \subseteq X$. The **inclusion function** of A in X is the function $\iota: A \rightarrow X$ (or $\iota: A \hookrightarrow X$) given by $\iota(x) = x$ for all $x \in A$.

4. If $f: X \rightarrow Y$ is a function and $A \subseteq X$, the **restriction** of f to A is the function $f|_A: A \rightarrow Y$ given by $f|_A(x) = f(x)$ for all $x \in A$.